

PROJECT REPORT

Import Data using Transform Maps (Spreadsheet)

1. INTRODUCTION

1.1 Project Overview

This micro project demonstrates the end-to-end import of structured employee data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The spreadsheet acts as the external source, the Employee Import table acts as a staging area, and the Employee Test table is the target. A Transform Map connects source fields to target fields and executes the transformation. The imported records are then validated, duplicate prevention is demonstrated with Coalesce, and reports are consolidated into an Employee Analytics Dashboard.

1.2 Purpose

The purpose is to provide a practical workflow for loading bulk employee information into ServiceNow accurately and efficiently. The project covers spreadsheet preparation, target-table creation, import-set loading, field mapping, transformation, validation, duplicate prevention, repeated imports, reporting, and dashboard creation.

2. IDEATION PHASE

2.1 Problem Statement

Employee information may arrive in spreadsheet form and can contain many records, changes to existing employees, and repeated submissions. Manually entering or repeatedly inserting such data can cause duplicate records and reduce data consistency. The project addresses this by staging spreadsheet data in an Import Set and transforming it into a controlled ServiceNow target table.

2.2 Empathy Map Canvas

Source-derived stakeholder view: administrators/HR users receive employee data externally, need accurate imports, need to identify inserted and updated records, and need visibility into employee information. Their operational concerns are duplicate records, unsuccessful imports, and manual effort. The implemented workflow responds through Import Sets, Transform Maps, Coalesce, Transform History, reports, and a dashboard.

2.3 Brainstorming

The solution flow developed from the source process is: prepare spreadsheet → create Employee Test target table → create Employee Import staging table → create Transform Map → auto-map and assist field mapping → transform → validate → enable Coalesce → import changed/new spreadsheet data → inspect Transform History → create reports → add reports to Employee Analytics Dashboards.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

A source-derived journey is: receive employee spreadsheet → prepare/download the spreadsheet → load data into Employee Import → configure source-to-target mappings → run the transformation → inspect Employee Test and Transform History → repeat imports when data changes → use Coalesce to update existing records rather than create duplicates → review reports and dashboard.

3.2 Solution Requirement

Functional requirements supported by the source are: create the Employee Test table (u_employee_test); create Employee ID, Employee Name, Email, Department and Location fields; create Employee Import (u_employee_import); create the Sample Spreadsheet Import Transform Map; map source and target fields; execute transformations; validate records; enable Coalesce; process new and changed spreadsheet rows; create Employees by Department, Employees by Location and Employee List Report; and add the reports to Employee Analytics Dashboards.

3.3 Data Flow Diagram

External spreadsheet data is loaded into the Employee Import staging table. The Transform Map maps the staged source fields to the Employee Test target table. The transformation creates or updates target records. Transform History provides import status information. The resulting employee data feeds reports and the Employee Analytics Dashboard. Coalesce provides duplicate-prevention/update behavior for repeated imports.

3.4 Technology Stack

ServiceNow platform; Import Sets / Load Data; Import Set table Employee Import (u_employee_import); target custom table Employee Test (u_employee_test); Transform Maps; Auto Map Matching Fields; Mapping Assist; Transform History; Coalesce; ServiceNow Reports; ServiceNow Dashboards; external spreadsheet/Excel source. The source does not specify external APIs, machine-learning components, or a separate application database.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem is bulk employee-data migration and repeated updates from spreadsheet files. The solution fits the workflow by using a staging table for incoming data, explicit field mappings for controlled transformation, a target employee table for operational records, Coalesce for duplicate prevention, and reports/dashboard components for visibility.

4.2 Proposed Solution

The proposed solution is an automated ServiceNow employee-data import workflow. A spreadsheet is prepared and downloaded as Sample Spreadsheet.xlsx. Data is loaded through Load Data into Employee Import. The Sample Spreadsheet Import Transform Map maps the source fields to Employee Test and transforms the records. Subsequent imports can update existing records or insert new records, while repeated identical submissions can be ignored when Coalesce identifies existing records. Reports and a dashboard provide centralized visibility.

4.3 Solution Architecture

Architecture layers: Spreadsheet/Excel source → Load Data / Import Set staging → Employee Import table → Sample Spreadsheet Import Transform Map → Employee Test target table → Transform History and validation → reports (Department, Location, List) → Employee Analytics Dashboard. Coalesce operates in the Transform Map to identify existing records during repeated imports.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work can be organized into two sprints based on the implementation flow. Sprint 1 covers spreadsheet preparation, Employee Test creation, Employee Import creation, field setup, and Transform Map mapping. Sprint 2 covers transformation and validation, Coalesce, repeated imports, reports, and dashboard creation.

The detailed planning logic used for this project totals 26 story points across two sprints, giving a calculated velocity of 13 story points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The source documents functional validation through successful transformation, Transform History, and repeated imports. In the demonstrated repeated-import scenario, four rows were uploaded: two new IDs were inserted and two existing IDs were updated with changed name/email values. When the same sheet was uploaded again, four rows were ignored, with zero new inserts and zero updates. These observations demonstrate the intended duplicate-prevention behavior. The source does not provide numerical latency, throughput, CPU, memory, or load-test measurements, so those performance metrics are not specified.

7. RESULTS

7.1 Output Screenshots

The source contains screenshots covering the Employee Import table, Transform Map, transformed Employee Test records, Transform History, report creation, dashboard creation, and the final dashboard with three reports. Representative source pages are included below as evidence of the implemented workflow.

8. ADVANTAGES & DISADVANTAGES

Advantages: reduces manual entry for bulk employee data; provides a staging layer; supports explicit source-to-target field mapping; validates transformed records; Coalesce prevents redundant duplicate records during repeated imports; Transform History shows import outcomes; reports and dashboards provide centralized visibility.

Disadvantages/limitations from the source: the workflow depends on correctly prepared spreadsheet data and field mappings; configuration requires ServiceNow administrative steps; the source does not document automated external-system integration, API integration, quantified performance testing, or a production-grade security design.

9. CONCLUSION

The project successfully implements an automated employee data management workflow in ServiceNow using Import Sets and Transform Maps. The process loads spreadsheet data into a staging table, maps it to the Employee Test target table, transforms and validates the records, and supports subsequent data updates. Coalesce improves data integrity by identifying existing records during repeated imports. Reports and the Employee Analytics Dashboard provide visual access to employee data and import activity.

10. FUTURE SCOPE

Based strictly on the demonstrated workflow, future extensions could include connecting the import process to additional external data sources, automating scheduled imports, adding more validation rules, expanding report metrics, introducing richer dashboard controls, and adding formal performance/security testing. These extensions are not implemented in the source document and are therefore presented as possible future work rather than completed features.

11. APPENDIX

Source Code (if any)

No source-code listing is provided in the supplied project document. The implementation is described through ServiceNow configuration and navigation steps.

Dataset Link

No external dataset link is provided. The source uses a sample Google Spreadsheet/Excel workflow and refers to Sample Spreadsheet.xlsx.

GitHub & Project Demo Link

No GitHub repository or project-demo URL is provided in the supplied source.

Representative Source Screenshots

Import Data using Transform Maps (Spreadsheet)

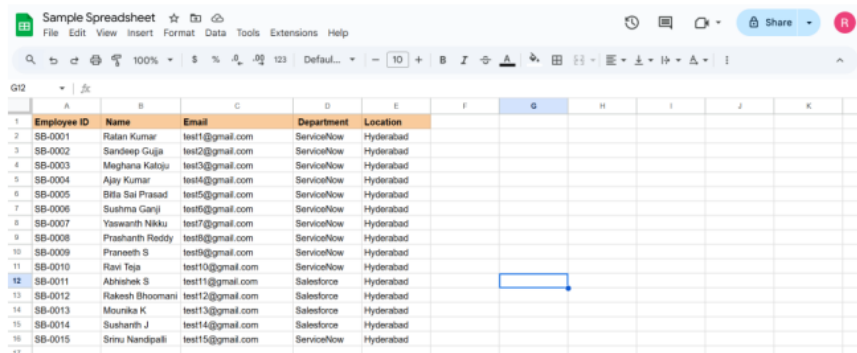
Description:

This micro project focuses on importing structured data from an external spreadsheet into the **ServiceNow** platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

Step by Step process:

Phase 1: Prepare the sheets

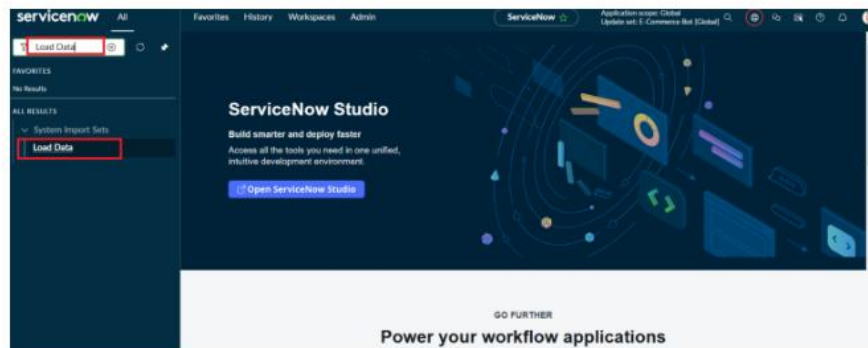
1. Open New Google Spreadsheet
2. Create New Google Spreadsheet with sample data as shown below



	A	B	C	D	E	F	G	H	I	J	K
1	Employee ID	Name	Email	Department	Location						
2	SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad						
3	SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad						
4	SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad						
5	SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad						
6	SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad						
7	SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad						
8	SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad						
9	SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad						
10	SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad						
11	SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad						
12	SB-0011	Alpesh S	test11@gmail.com	Salesforce	Hyderabad						
13	SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad						
14	SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad						
15	SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad						
16	SB-0015	Srinu Nandipati	test15@gmail.com	ServiceNow	Hyderabad						
17											

3. Download the spreadsheet as Sample Spreadsheet.xlsx format.

Figure 1. Source workflow screenshot (page 1).



Click on Load Data and create an import table
Fill details as shown below

- Label: Employee Import
- Name: u_employee_import (auto populated)

- Click on Submit

Once we click on Submit, we can see the success message as shown below.

Also we need to make sure to click on Create Transform Map.

Figure 2. Source workflow screenshot (page 5).

Progress

Name	ImportProcessor
State	Complete
Completion code	Success
Message	Processed: 15, inserts: 15, updates: 0, errors: 0, empty and ignored: 0, ignored errors: 0 (0/0/0/1,348)

Next steps...

Import sets - Go to the import sets for this data load

Loaded data - Go to the newly imported data inside the staging table: u_employee_import

Create transform map - Create a transform map for the newly staged data

Import log - View the import log

Phase 4: Create Transform Map

In this phase we need to create the Transform Map by clicking on the Create Transform Map button as shown in above screenshot.

Give inputs as shown below.

Name: Sample Spreadsheet Import

Source Table: Employee Import is auto-populated

Target Table: Employee Test - as we created earlier

Click on Auto Map Matching fields.

Table Transform Map

New record

Name: Sample Spreadsheet Import

Source table: Employee Import (u_employee_import)

Active: ☒

Run business rules: ☒

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty candidate fields: ☐

Application: Global

Created: [Date]

Target table: Employee Test (u_employee_test)

Order: 100

Run script: ☐

Related Links

[Auto Map Matching Fields](#)

Now click on Mapping Assist and map the fields with Source & Target Table

Mapping Assist

Source: Employee Import

Target: Employee Test

Name	Employee Name
Employee ID	Employee ID
Department	Department
Location	Location
Email	Email

Save Cancel

Click on Save

Make sure the both Source & Target table fields are mapped as shown below.

Figure 3. Source workflow screenshot (page 6).

Progress	
Name	Transforming_HST0010001
State	Complete
Completion code	Success
Message	ST0010001 complete

Next steps...

[HST0010001](#) Go to the import sets for this data load

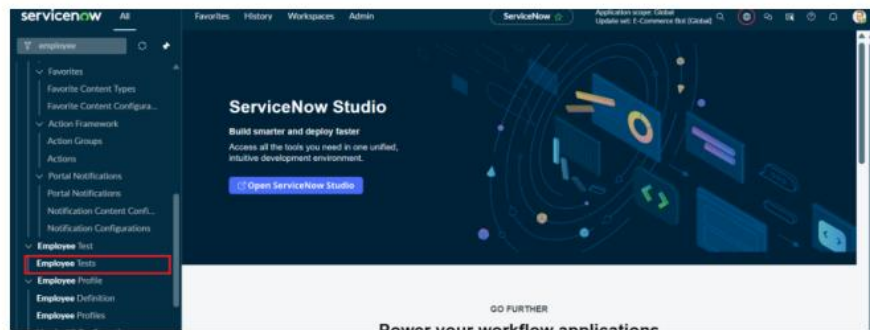
[Transform history](#) Show the transform history, related errors and log

[Import log](#) View the import log

Phase 5: Transform Data & Validate

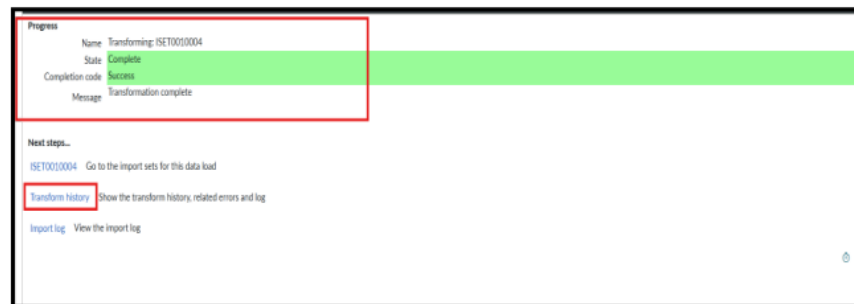
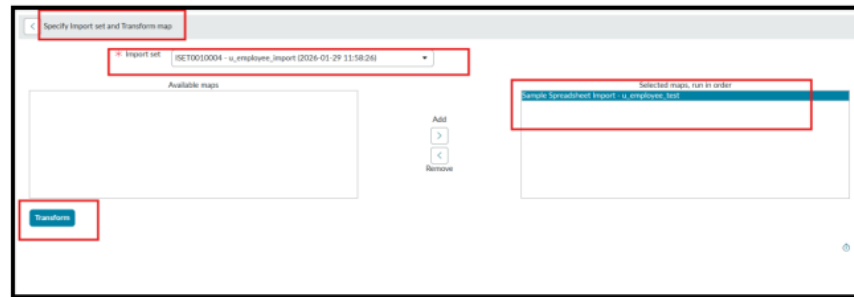
Navigation:

- Open Employee Test Table from application navigator as shown below.



- As our import got succeeded, all the records are created in the Employee Test Table.

Figure 4. Source workflow screenshot (page 8).



- Now the State is completed and Success.
- Now again click on the Transform history then it shows the newly added data and its status of Inserted and Skipped values out the given data.

Transform Histories										
Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped
2026-01-29 03:47:37	Complete	2026-01-29 03:47:38	1 Second	ISET0010003	Employee Import (u_employee_import)	4	2	2	0	0

Figure 5. Source workflow screenshot (page 15).

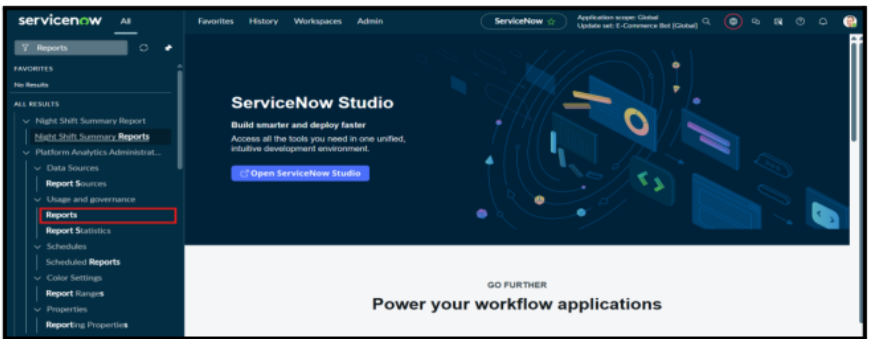
PHASE 8 : Create Reports on Employee Test Table

Navigation Steps :

- Open All menus, Search for **Reports**.
- Under **Usage and governance** Select Reports.
- Click the **New** button to create a new report.
- Create **3 Reports** as follows.

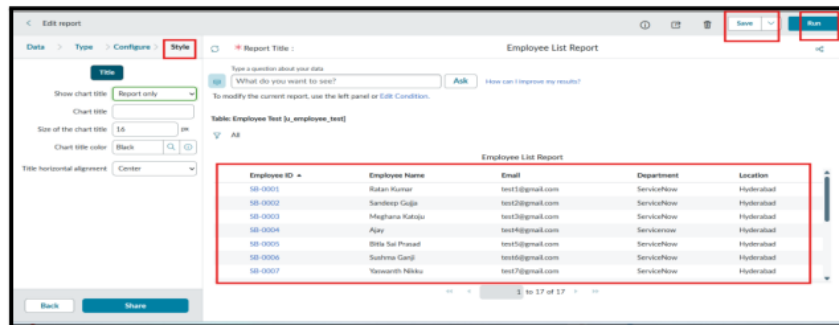
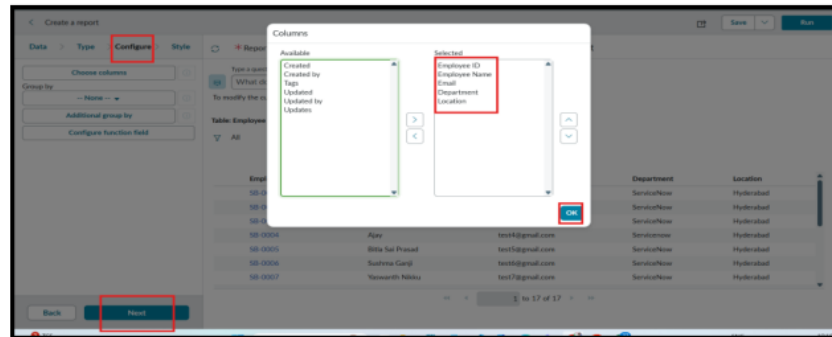
Report 1: Employees by Department

- In Data
- Report name : **Employees by Department**.
- Source Type : **Table**.
- Table : **Employee Test**, Now click the Next button.



Reports						
Title	Table	Type	Field Name	Created by	Updated	
SLA Performance Trend	Incident [incident]	Bar	opened_at	admin	2026-01-02 02:32:06	
Pending Justifications	Incident [incident]	List		admin	2026-01-02 01:13:00	
Top Assignees with SLA Violations	Incident [incident]	Bar	assigned_to	admin	2026-01-01 22:54:58	
Acknowledged vs Unacknowledged Breaches	Incident [incident]	Pie	u_sla_acknowledged	admin	2026-01-01 22:47:18	
SLA Breaches by Assignment Group	Incident [incident]	Bar	assignment_group	admin	2026-01-01 22:23:49	
SLA Breaches by Reason	Incident [incident]	Pie	u_sla_breach_reason	admin	2026-01-01 22:15:34	
Compliant CI Trend	CMDB Health Scorecard [cmdb_health_scorecard]	Line		admin	2025-12-17 05:52:39	

Figure 6. Source workflow screenshot (page 17).



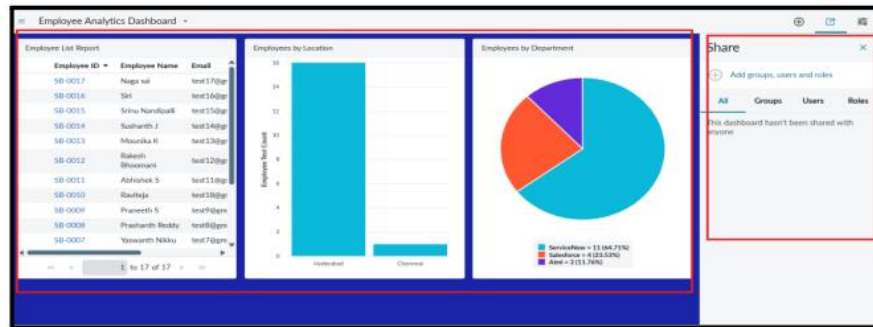
PHASE 9 : Add Reports to Dashboard

Create Dashboard

Navigation :

- Open All menus → Search for Dashboards.
- Under Self Service → Click Dashboards.
- On the top right corner Click Create Dashboards or /
- You can create Directly by using the backend name [PA_DASHBOARDS.FORM].
- Then Click New to Create new Dashboards.

Figure 7. Source workflow screenshot (page 24).



- Here the three Reports are added successfully by varying **Department , Location and List Report**.
- We can also restrict the **Users,Groups and Roles** From Share Option.

Conclusion:

1. This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.
2. A key enhancement in this project was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system intelligently identifies existing employee records during every import. This prevents the creation of duplicate records and ensures that repeated imports update existing data instead of inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.
3. In addition to data integrity, the project was extended with Reports and Dashboards to provide meaningful insights and real-time visibility into employee data and import activities. Custom reports were created to track:

Figure 8. Source workflow screenshot (page 30).

Source basis: Import Data using Transform Maps (Spreadsheet), supplied as the project source document. The report follows the section order of the uploaded Final Report Template.